air capture (DAC), point source fossil (PSF) and point source biogenic (PSB)), and Others (including mixture of sources).
3. Source of electricity/energy
Sustainable marine fuels requiring significant electricity input during WtT phase and electricity delivered directly to ships are produced by using electricity/energy from renewable, nuclear or biogenic sources, which are additional to current or long-standing demand levels, or by using surplus electricity during off-peak hours.
1. The GHG intensity of electricity used in the production of marine fuels or delivered directly to ships (annual average, expressed in g CO2eq/kWh based on total emissions and actual hours of production.
4. Carbon stock direct land use change (DLUC)

Sustainable marine fuels are not made from biomass obtained from land with high carbon stock; production of sustainable marine fuels minimizes emissions resulting from Direct Land Use Change.
1. Sustainable marine fuel feedstock does not include biomass obtained from land with high

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SOURCE: MEPC.376(80).pdf
obtained from land with high carbon stock; production of sustainable marine fuels minimizes emissions resulting from Direct Land Use Change.
1. Sustainable marine fuel

feedstock does not include biomass obtained from land with high carbon stock (e.g. primary forests, wetlands, or peat lands referred to a specific cut-off date for conversion), or a sustainable land management plan and reporting schedule are in

14 Pending further guidance to be developed by the Organization.

Theme/aspect Principle/Objective Metric/Indicator

place to ensure that the biomass is obtained from activities or ecosystem services that do not negatively impact the soil carbon stock;
2. The production of sustainable marine fuels does not occur in lands converted from primary forest, forestland, grassland or legally protected land, taking (1 January 2008)14 as the cut-off date; and
3. Direct land-use change (DLUC) indicator, expressed in GHG (including CO2, CH4 and

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SOURCE: MEPC.376(80).pdf

converted from primary forest, forestland, grassland or legally protected land, taking (1 January 2008)14 as the cut-off date; and
3. Direct land-use change (DLUC) indicator, expressed in GHG (including CO2, CH4 and
N2O emissions) intensity,

i.e. mass of CO2 equivalent / MJ of production or yield of feedstock.
5. Carbon stock indirect land use change (ILUC)

Cultivation of feedstock of sustainable marine fuels minimizes inducing negative changes in the use or management of land which occurs outside the product system being assessed.
1. Indirect carbon stock risk associated with cultivation of feedstock for sustainable marine fuels (see para. 4.13).
6. Water

Production of sustainable marine fuels maintain or enhance water quality and availability.
1. Operational practices are in place to (1) maintain water quality; and (2) use water efficiently and to avoid the depletion of water resources (including surface water, renewable water and fossil/underground water) beyond replenishment capacities;
2. Respect of decision-making of local population on water management;
3. Water environment impact (weighted water consumption on water

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SOURCE: MEPC.376(80).pdf
surface water, renewable water and fossil/underground water) beyond replenishment capacities;
2. Respect of decision-making of local population on water management;

3. Water environment impact
(weighted water
consumption on water
scarcity);

Theme/aspect Principle/Objective Metric/Indicator
4. Water Use Indicator
expressed in m3/year per
MJ or production or yield of
feedstock;
5. Freshwater eutrophication
indicator, e.g. expressed in
kg of phosphorus
equivalent (Peq) and kg of
nitrogen equivalent (Neq)
released to fresh water/kg
of feedstock produced or
per MJ respectively; and